

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence
CODE: CSA17

COURSE

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Hashini S / 192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Hashini S

1. Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Problem

Obtain exactly **2 gallons** in the **4-gallon jug**.

State Representation

State = (4-gallon jug, 3-gallon jug)

Initial State:

(0,0)

Goal State:

(2,x) where x can be 0–3.

Steps

Step	Action	State (4G,3G)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0) ✓

Goal Achieved: Exactly **2 gallons** are in the 4-gallon jug.

2. Mars Rover Intelligent Agent

i) Percepts

- Camera images
- Rock and soil characteristics
- Temperature
- Atmospheric pressure
- Terrain obstacles
- Radiation level
- GPS/position data

- Wheel status
- Battery level

ii) Operating Environment

- **Partially Observable**
- **Stochastic**
- **Sequential**
- **Dynamic**
- **Continuous**
- **Single Agent**

iii) Actions

- Move forward/backward
- Turn left/right
- Capture images
- Collect soil samples
- Analyze rocks
- Drill surface
- Send data to Earth
- Recharge using solar panels
- Avoid obstacles

iv) Performance Measure

- Number of successful samples collected
- Distance covered
- Accuracy of scientific observations
- Energy efficiency
- Safe navigation
- Mission completion rate

v) Suitable Agent Architecture

Utility-Based Learning Agent

Reason

- Makes decisions based on maximizing scientific value.
- Learns from previous observations.
- Adapts to unknown environments.
- Optimizes battery usage and route planning.

3. 8-Queens Problem Formulation

Initial State

Empty chessboard with no queens.

State Space

All possible arrangements of queens on the board.

Operators (Actions)

Place one queen in a safe position of the next column.

Goal Test

- Eight queens placed.
- No two queens attack each other.

That means:

- No same row
- No same column
- No same diagonal

Path Cost

Uniform cost = 1 per queen placement.

Search Strategies

- Breadth First Search (BFS)
- Depth First Search (DFS)
- Backtracking (Most Efficient)
- Hill Climbing
- A* Search (less common)

4. OLA Cab Booking as a Problem-Solving Agent

Type of Agent

Goal-Based Problem Solving Agent

Reason

The customer's goal is to reach the destination with suitable comfort and minimum travel time/cost.

Pseudocode

Algorithm OLA_CAB_BOOKING

Input:

Source

Destination

Available Cab Types

Begin

Enter Source

Enter Destination

Display available cab types

Select cab

Estimate fare

Estimate arrival time

If cab available then

 Confirm booking

 Assign nearest driver

 Driver reaches customer

 Customer travels to destination

 Payment completed

Else

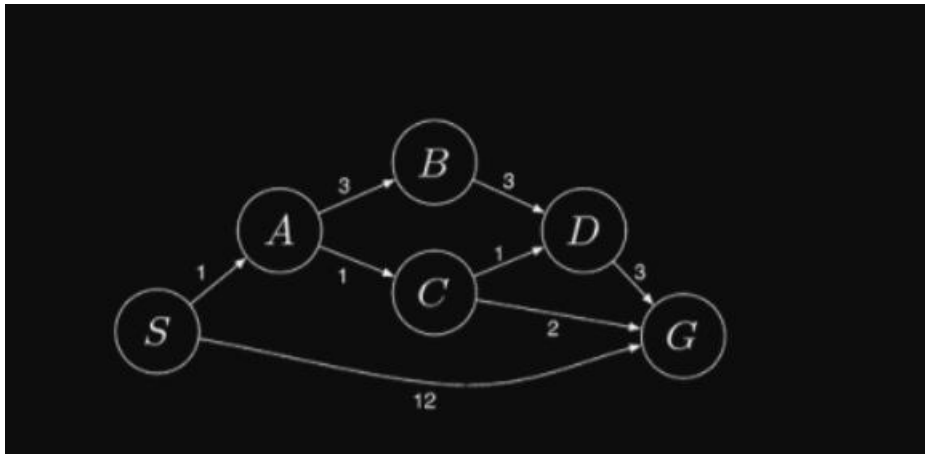
 Display "Cab Not Available"

End If

End

5. Uniform Cost Search (UCS) – Logistics Delivery Network

For this graph, apply **Uniform Cost Search (UCS)** by always expanding the node with the **lowest cumulative path cost**.



Step 1: Graph Details

Edges and costs:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform Cost Search (UCS)

Initial State

- Frontier = $\{(S, 0)\}$
- Visited = $\{\}$

Iteration 1

Expand S (**cost = 0**)

Add successors:

- A (1)

- G (12)

Frontier:

Node Cost

A 1

G 12

Iteration 2

Expand A (**cost = 1**)

Add successors:

- $B = 1 + 3 = 4$

- $C = 1 + 1 = 2$

Frontier:

Node Cost

C 2

B 4

G 12

Iteration 3

Expand C (**cost = 2**)

Add successors:

- $D = 2 + 1 = 3$

- $G = 2 + 2 = 4$

Frontier:

Node Cost

D 3

B 4

G 4

Node Cost

G 12

Iteration 4

Expand **D** (cost = 3)

Add successor:

- $G = 3 + 3 = 6$

Frontier:

Node Cost

B 4

G 4

G 6

G 12

Iteration 5

Expand **B** (cost = 4)

Add successor:

- $D = 4 + 3 = 7$ (ignored because D was already reached with lower cost 3)

Frontier:

Node Cost

G 4

G 6

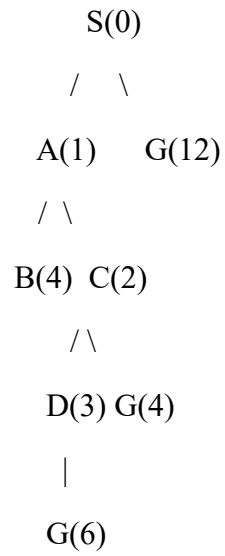
G 12

Iteration 6

Expand **G** (cost = 4)

Goal reached.

Search Tree



Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$S \rightarrow A = 1$

$A \rightarrow C = 1$

$C \rightarrow G = 2$

Total Cost = 4

Final Answer

- **Algorithm Used:** Uniform Cost Search (UCS)
- **Least-Cost Path:** $S \rightarrow A \rightarrow C \rightarrow G$
- **Minimum Path Cost:** 4